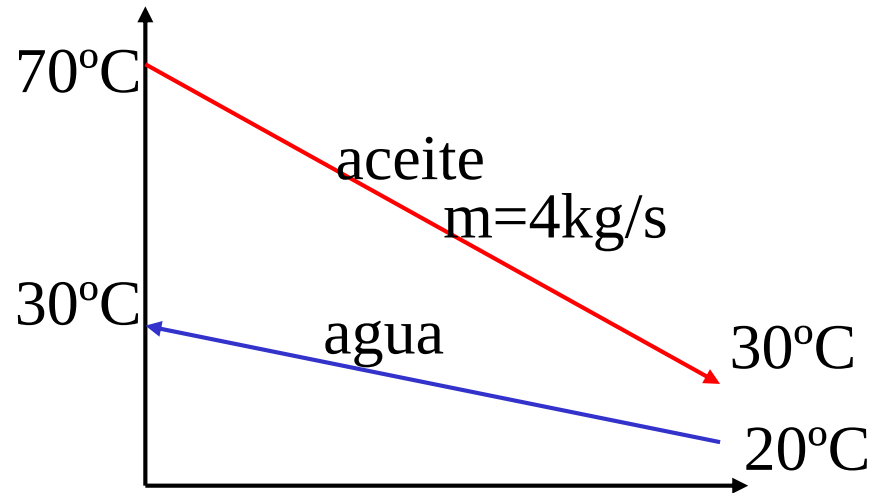


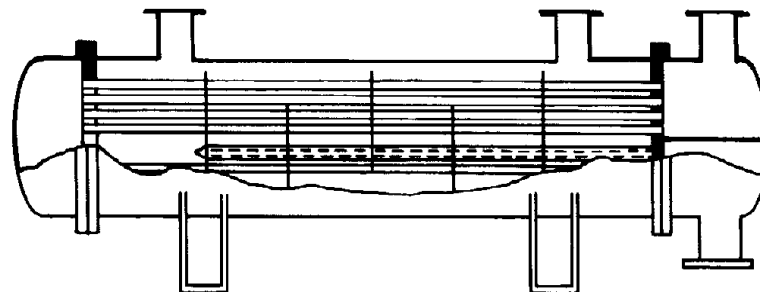
- Cálculo de áreas y temperaturas de salida
- Selección de intercambiadores
- Configuraciones en serie de tubo-carcasa

Determinar el área de transferencia de un intercambiador tubo carcasa tipo E con un n° par de pasos por los tubos para un intercambio aceite agua dadas las temperaturas y caudales propuestos



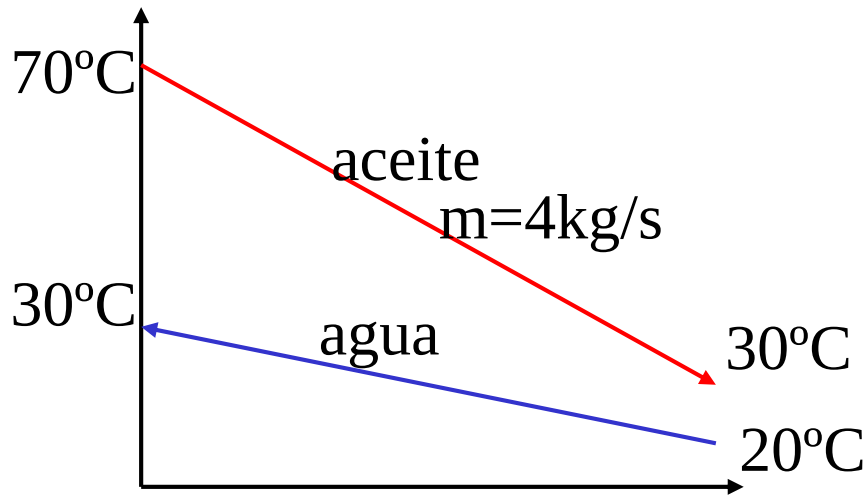
$$c_{p,c} = 4200 \text{ J/kgK} \quad A$$

$$c_{p,h} = 2000 \text{ J/kgK}$$



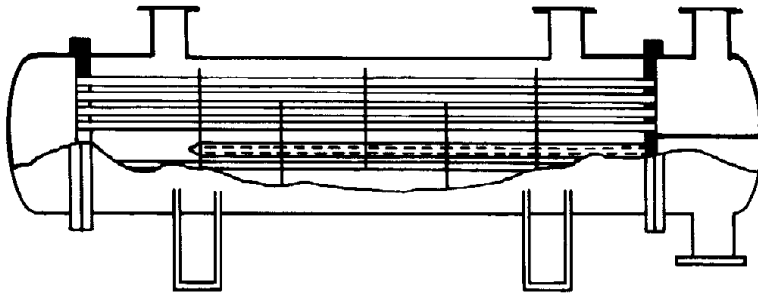
$$U = 700 \text{ W/m}^2\text{K}$$

Determinar el área del intercambiador

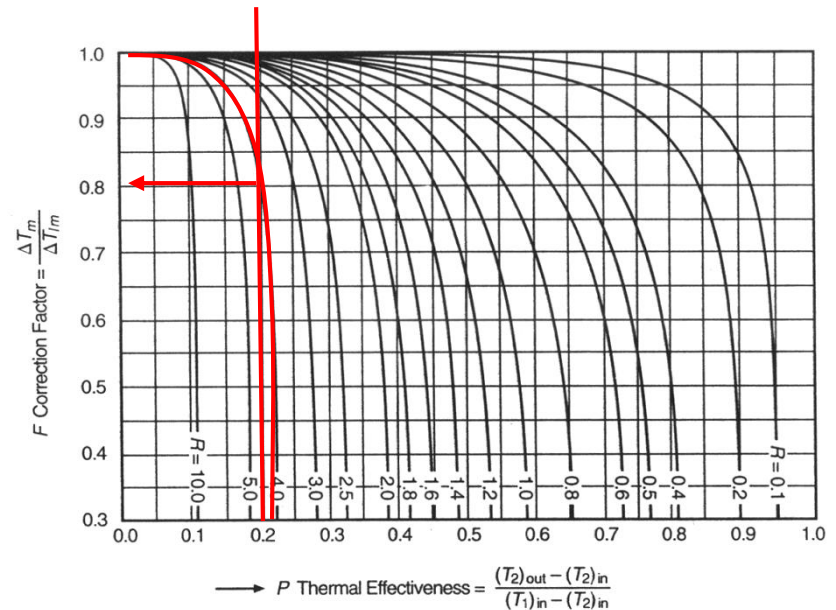
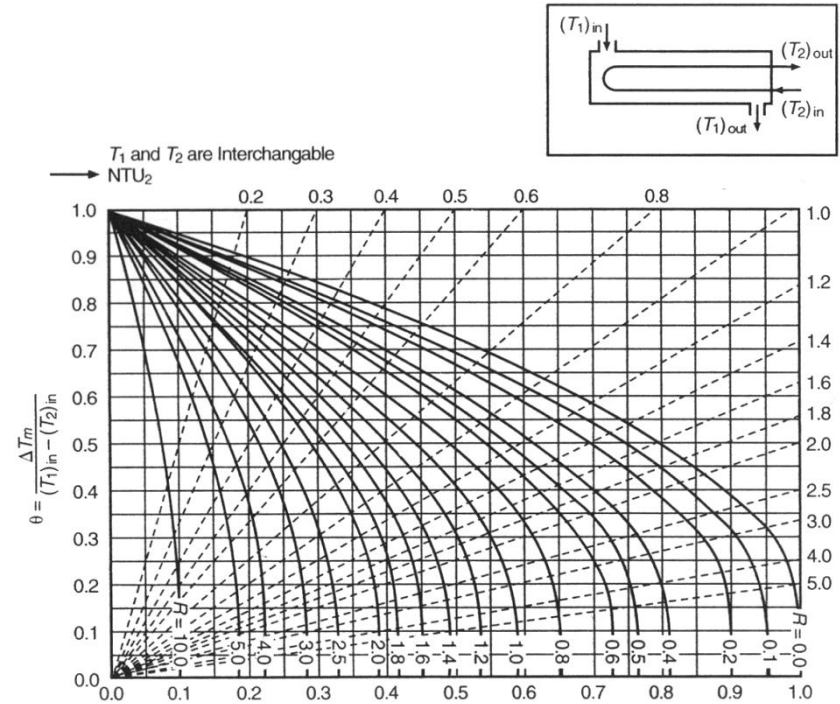


$$C_{p,c} = 4200\text{J/kgK} \quad A$$

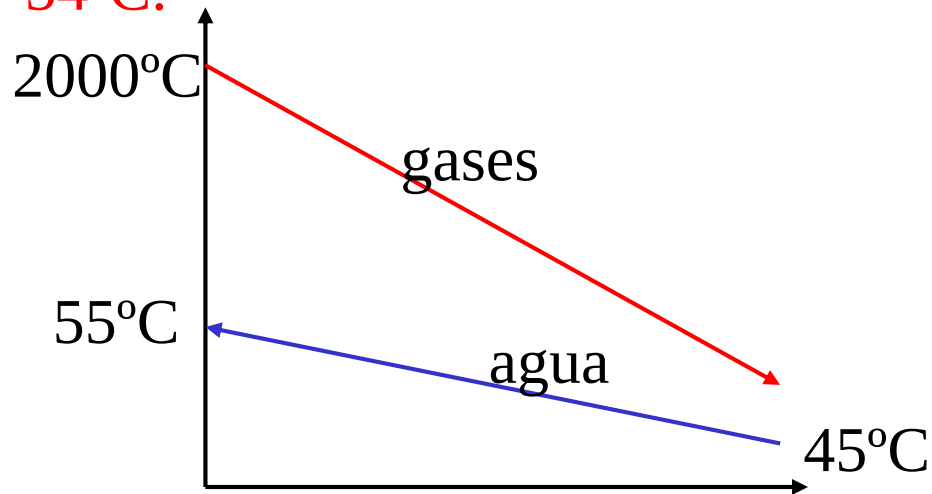
$$C_{p,h} = 2000\text{J/kgK}$$



$$U = 700\text{W/m}^2\text{K}$$

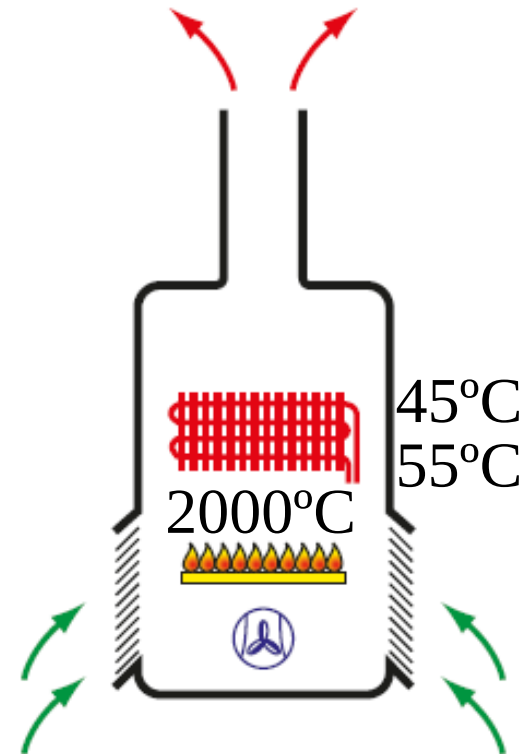


Determinar el intercambiador de una caldera de 30kW, lo más eficiente posible, sabiendo que los gases entran al intercambiador a 2000°C y que el hay que calentarla de 45°C a 55°C. Para determinar el área sabemos que el coeficiente local, del lado de los humos es de 50 W/m²K y el global 40 W/m²K y que se debe evitar que se produzca condensación ácida, siendo la temperatura de condensación ácida 54°C.



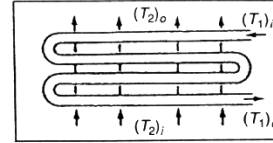
$$U = 40 \text{ W/m}^2\text{K}$$

$$h = 50 \text{ W/m}^2\text{K}$$



$$50(T_{go} - 54) = 40(T_{go} - 45) \Rightarrow T_{go} = 90^{\circ}\text{C}$$

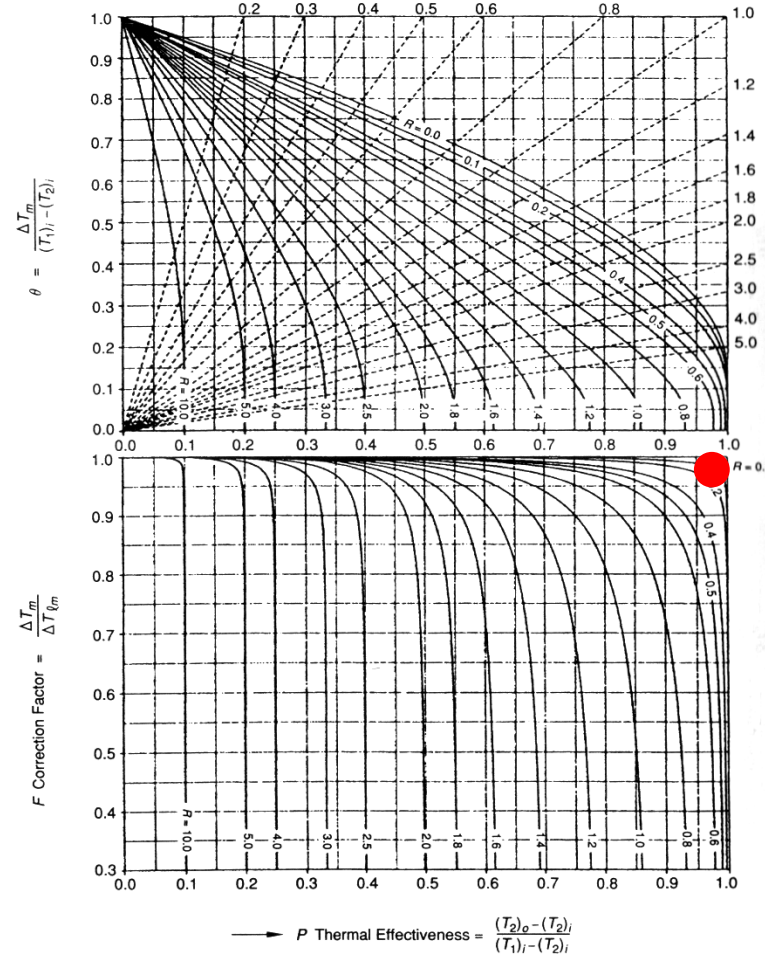
$$\Delta T_{lm} = 504.5^{\circ}\text{C}$$



$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

T_1 and T_2 are not interchangeable

$$\text{NTU}_2 = AU/C_2$$

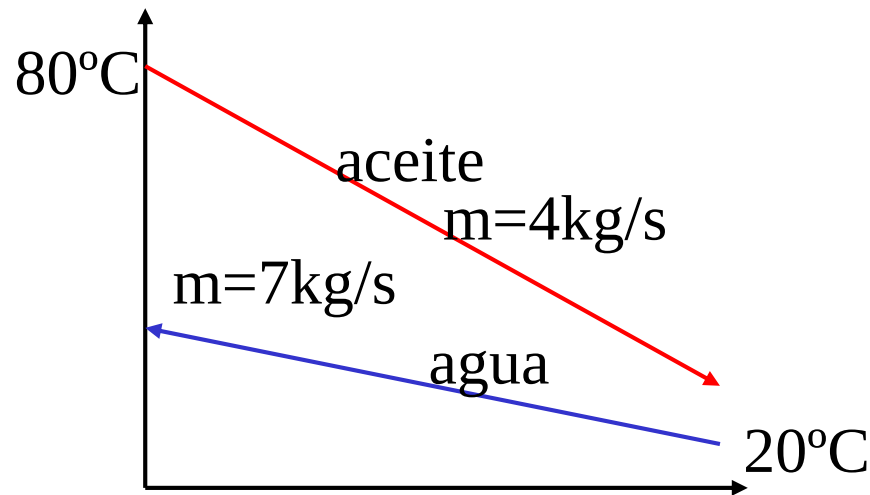


$$P = \frac{90 - 2000}{45 - 2000} = 0.98$$

$$R = \frac{45 - 55}{90 - 2000} = 0.005$$

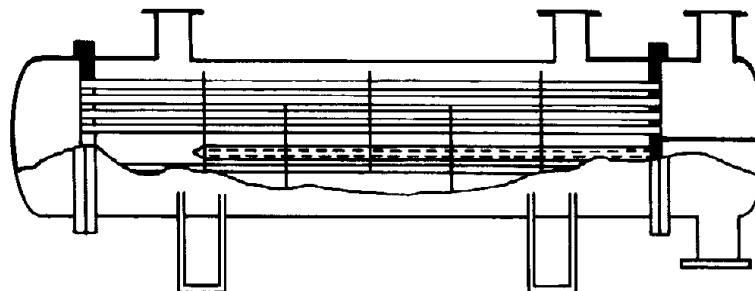
$$A = \frac{30000}{40 \cdot 504.5} = 1.5 \text{m}^2$$

Determinar las temperatura de salida de un intercambiador tubo carcasa tipo E con un n° par de pasos por los tubos de 100 m^2 y $U = 700 \text{ W/m}^2\text{K}$ conocidas las temperaturas de entrada y los caudales



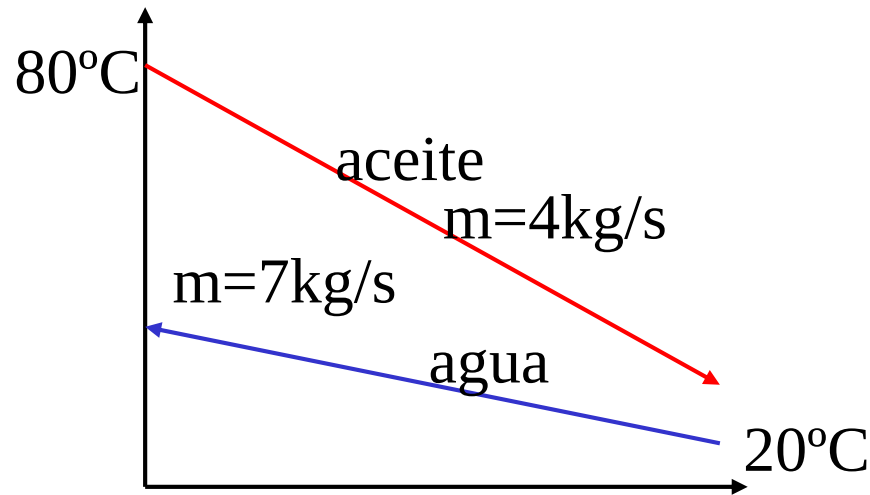
$$c_{p,c} = 4200 \text{ J/kgK} \quad A$$

$$c_{p,h} = 2000 \text{ J/kgK}$$



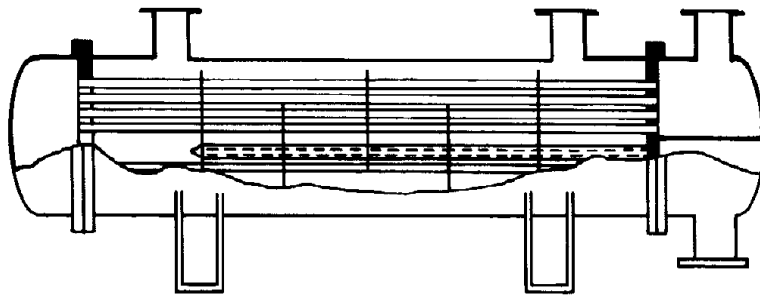
$$U = 700 \text{ W/m}^2\text{K}$$

$$A = 100 \text{ m}^2$$



$$C_{p,c} = 4200 \text{ J/kgK} \quad A$$

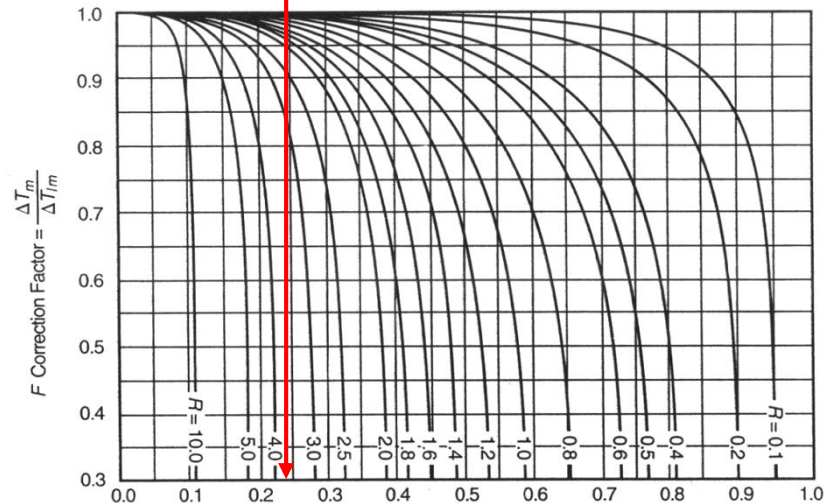
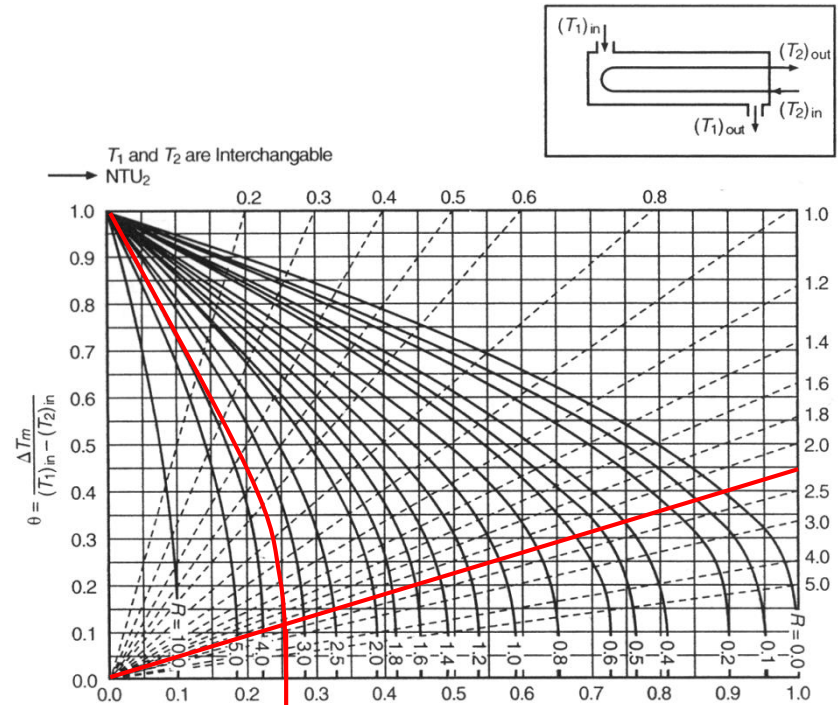
$$C_{p,h} = 2000 \text{ J/kgK}$$



$$U = 700 \text{ W/m}^2\text{K}$$

$$A = 100 \text{ m}^2$$

$$UA = 70000 \text{ W/K}$$



$$P \text{ Thermal Effectiveness} = \frac{(T_2)_{out} - (T_2)_{in}}{(T_1)_{in} - (T_2)_{in}}$$

Determinación de U

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 5.70	93 5.02	120 5.51	102 4.93	99 4.96	63 5.50	107 4.87	100 4.95	86 5.11
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 5.02	300 4.18	350 4.81	429 4.03	375 4.09	120 4.76	530 3.95	388 4.07	240 4.28
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 5.51	350 4.81	400 6.25	600 4.56	450 4.38	200 5.50	600 4.56	400 4.82	300 4.81
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 4.89	484 3.98	600 4.56	938 3.77	714 3.85	142 4.59	1607 3.61	764 3.83	345 4.12
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 5.39	138 4.61	200 5.50	161 4.46	153 4.51	82 5.16	173 4.42	155 4.50	214 4.33
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 4.89	467 3.99	550 4.91	875 3.79	677 3.87	140 4.60	1432 3.64	722 3.85	336 4.13
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
5 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 2.11	93 1.63	120 2.26	102 1.58	99 1.59	63 1.95	107 1.55	100 1.59	86 1.68
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 1.63	300 1.11	350 1.89	429 1.02	375 1.05	120 1.49	530 0.98	388 1.05	240 1.18
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 2.26	350 1.89	400 2.25	600 1.10	450 1.46	200 1.93	600 1.10	400 1.45	300 1.45
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 1.56	484 1.00	600 1.10	938 0.88	720 0.91	142 1.41	1607 0.83	764 0.90	345 1.07
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 1.59	375 1.05	450 1.46	600 0.95	500 0.99	130 1.46	818 0.89	524 0.98	286 1.13
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 1.86	138 1.43	200 1.93	161 1.36	153 1.38	82 1.71	173 1.32	155 1.37	214 1.48
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 1.56	467 1.00	550 1.20	875 0.88	677 0.93	140 1.42	1432 0.84	722 0.91	336 1.08
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 1.59	375 1.05	450 1.46	600 0.95	500 0.99	130 1.46	818 0.89	524 0.98	286 1.13

Seleccionar el intercambiador adecuado para las características de operación que se proponen en el cuadro adjunto

1	Fluido	C_p (kJ/kgK)	m (kg/s)	P (bar)	T_{in} (°C)	T_{out} (°C)	Características
Hot	Aceite	2	0.5	30	70	30	-Altamente viscoso. -Ensuciamiento medio.
Cold	Agua	4.2	2		20		

1	Fluido	C_p (kJ/kgK)	m (kg/s)	P (bar)	T_{in} (°C)	T_{out} (°C)	Características
Hot	Aceite	2	0.5	30	70	30	-Altamente viscoso. -Ensuciamiento medio.
Cold	Agua	4.2	2		20		

Heat exchanger type	Maximum pressure (absolute)	Temperature range	Fluid limitation	Normal size ranges for individual unit	Special features
Air-cooled	High on process side (500 bar).	High temperature possible on process side (up to 600°C).	Subject only to material of construction.	5 to 350 m ² (bare tube) (per bundle – multiple bundles often used).	Heat rejection system. Requires fans. Tubes normally finned.
Brazed-plate	16 bar	Up to 200°C	Suitable for liquids compatible with braze material.	1 to 10 m ²	Modular construction. Not easily cleaned.
Double-pipe (plain and finned tubes)	300 bar (shell) (higher pressures possible with special design). 1400 bar (tube).	-100 to 600°C (higher with special materials).	Subject only to materials of construction.	0.25 to 200 m ² per unit – multiple units are often used.	High thermal efficiency, standard modular construction.
Heat-pipe	Near atmospheric.	<200°C (higher with special heat pipe fluids).	Low pressure gases.	100 to 1000 m ²	Near counter-flow operation possible. Extended surface possible on both sides.
Plate-and-frame	25 bar with some models up to 40 bar.	-25 to 175°C (-40 to 200°C possible for very special types)	Normally unsuitable for gases. Limitation is on gaskets	1 to 2500 m ²	Modular construction. Normally the most economical if applicable.
Plate-fin	100 bar (aluminium). 200 bar (stainless steel).	-273 to +150°C (aluminium) (up to 600°C in stainless steel).	Low fouling.	<9m ³ volume	Very small possible. Incorporation of multiple streams. Very large surface area per unit volume.
Shell-and-tube	300 bar (shell). 1400 bar (tube).	-25 to 600°C (lower and higher with special materials).	Subject only to materials of construction.	10 to 1000 m ² (per shell – multiple shells can be used).	Very adaptable and can be used for nearly all applications.
Welded-plate	60 bar	In excess of 650°C	Subject only to materials of construction. Not suitable for fouling duties.	>1000 m ²	Differential pressure should be less than 30 bar. Differential expansion should be borne in mind.

$$\Delta T_{lm} = 23.35^\circ C$$

$$R = 0.119$$

$$P = 0.8$$

$$\frac{Q}{\Delta T_m} = \frac{Q}{F \Delta T_{lm}} = 1823 W/K$$

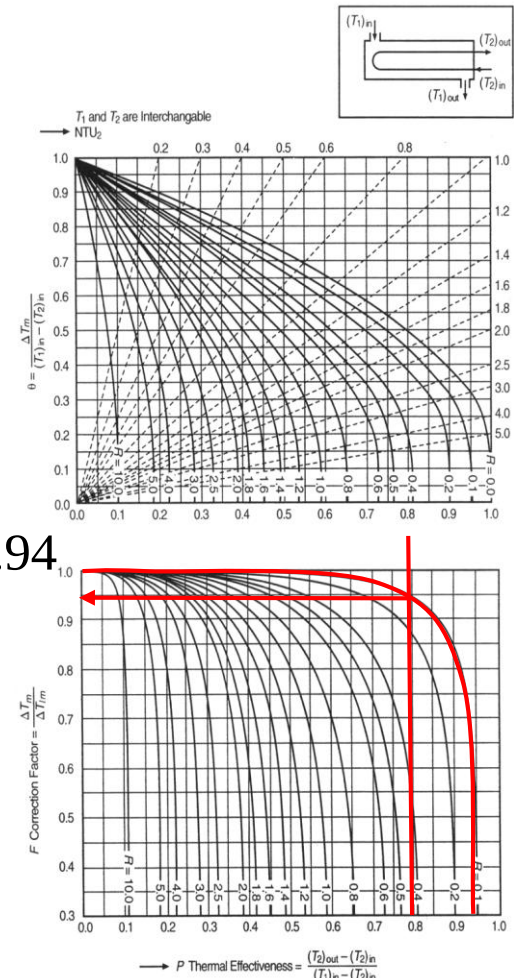
$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	$U (W/m^2 K)$ $C (\$/ (W/K))$	55 5.70	93 5.02	120 5.51	102 4.93	99 4.96	63 5.50	107 4.87	100 4.95	86 5.11
	Medium Pressure Gas (20 bar)	$U (W/m^2 K)$ $C (\$/ (W/K))$	93 5.02	300 4.18	350 4.81	429 4.03	375 4.09	120 4.76	530 3.95	388 4.07	240 4.28
	High Pressure Gas (150 bar)	$U (W/m^2 K)$ $C (\$/ (W/K))$	120 5.51	350 4.81	400 6.25	600 4.56	450 4.38	200 5.50	600 4.56	400 4.82	300 4.81
	Treated Cooling Water	$U (W/m^2 K)$ $C (\$/ (W/K))$	105 4.89	484 3.98	600 4.56	938 3.77	714 3.85	142 4.59	1607 3.61	764 3.83	345 4.12
	Low Viscosity Organic Liquid	$U (W/m^2 K)$ $C (\$/ (W/K))$	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
	High Viscosity Liquid	$U (W/m^2 K)$ $C (\$/ (W/K))$	68 5.39	138 4.61	200 5.50	161 4.46	153 4.51	82 5.16	173 4.42	155 4.50	214 4.33
	Boiling Water	$U (W/m^2 K)$ $C (\$/ (W/K))$	105 4.89	467 3.99	550 4.91	875 3.79	677 3.87	140 4.60	1432 3.64	722 3.85	336 4.13
5 000	Boiling Organic Liquid	$U (W/m^2 K)$ $C (\$/ (W/K))$	99 4.96	375 4.09	450 4.38	600 3.91	500 3.97	130 4.67	818 3.81	524 3.95	286 4.20
	Low Pressure Gas (< 1 bar)	$U (W/m^2 K)$ $C (\$/ (W/K))$	55 2.11	93 1.63	120 2.26	102 1.58	99 1.59	63 1.95	107 1.55	100 1.59	86 1.68
	Medium Pressure Gas (20 bar)	$U (W/m^2 K)$ $C (\$/ (W/K))$	93 1.63	300 1.11	350 1.89	429 1.02	375 1.05	120 1.49	530 0.98	388 1.05	240 1.18
	High Pressure Gas (150 bar)	$U (W/m^2 K)$ $C (\$/ (W/K))$	120 2.26	350 1.89	400 2.25	600 1.10	450 1.46	200 1.93	600 1.10	400 1.45	300 1.45
	Treated Cooling Water	$U (W/m^2 K)$ $C (\$/ (W/K))$	105 1.56	484 1.00	600 1.10	938 0.88	720 0.91	142 1.41	1607 0.83	764 0.90	345 1.07
	Low Viscosity Organic Liquid	$U (W/m^2 K)$ $C (\$/ (W/K))$	99 1.59	375 1.05	450 1.46	600 0.95	500 0.99	130 1.46	818 0.89	524 0.98	286 1.13
	High Viscosity Liquid	$U (W/m^2 K)$ $C (\$/ (W/K))$	68 1.86	138 1.43	200 1.93	161 1.36	153 1.38	82 1.71	173 1.32	155 1.37	214 1.48
	Boiling Water	$U (W/m^2 K)$ $C (\$/ (W/K))$	105 1.56	467 1.00	550 1.20	875 0.88	677 0.93	140 1.42	1432 0.84	722 0.91	336 1.08
	Boiling Organic Liquid	$U (W/m^2 K)$ $C (\$/ (W/K))$	99 1.59	375 1.05	450 1.46	600 0.95	500 0.99	130 1.46	818 0.89	524 0.98	286 1.13

$$C_1 = 4.59; (Q/\Delta T_m)_1 = 1000$$

$$C = 2.955$$

$$C_2 = 1.41; (Q/\Delta T_m)_2 = 5000$$

$$\text{Coste tubo-carcasa} = 2.955 \cdot 1823 = 5387 \text{€}$$



$$F = 0.94$$

$$\Delta T_{lm} = 23.35^\circ C$$

$$\frac{Q}{\Delta T_m} = \frac{Q}{\Delta T_{lm}} = 1713 W/K$$

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Hydrocarbon Liquid	High Viscosity Hydrocarbon Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
1 000	Low Pressure Gas (< 1 bar)	$U (W/m^2 K)$ $C (£/(W/K))$	55 4.8	95 3.8	125 2.9	105 3.8	100 3.8	65 4.7	110 2.8	100 3.8	85 3.9
	Medium Pressure Gas (20 bar)	$U (W/m^2 K)$ $C (£/(W/K))$	95 3.4	300 2.5	350 2.9	430 2.5	375 5	120 2.6	530 2.5	390 2.5	240 2.5
	High Pressure Gas (150 bar)	$U (W/m^2 K)$ $C (£/(W/K))$	120 2.9	350 2.9	400 2.9	500 2.9	400 2.9	150 2.9	600 2.9	420 2.9	350 2.9
	Treated Cooling Water	$U (W/m^2 K)$ $C (£/(W/K))$	105 2.8	484 2.5	500 2.9	940 2.5	715 2.5	145 2.5	1610 2.5	765 3.9	345 2.5
	Low Viscosity Hydrocarbon Liquid	$U (W/m^2 K)$ $C (£/(W/K))$	100 2.7	375 2.5	425 2.9	600 2.5	500 2.5	130 2.5	820 2.5	525 2.5	290 2.5
	High Viscosity Hydrocarbon Liquid	$U (W/m^2 K)$ $C (£/(W/K))$	70 4.7	140 2.5	175 2.9	160 2.5	155 2.5	85 3.9	175 2.5	155 2.5	125 2.5
	Boiling Water	$U (W/m^2 K)$ $C (£/(W/K))$	105 3.9	470 2.5	550 2.9	875 2.5	670 2.5	140 2.5	1435 2.5	725 2.5	340 2.5
	Boiling Organic Liquid	$U (W/m^2 K)$ $C (£/(W/K))$	100 4	375 2.5	430 2.9	600 2.5	500 2.5	130 2.5	820 2.5	525 2.5	285 2.5
5 000	Low Pressure Gas (< 1 bar)	$U (W/m^2 K)$ $C (£/(W/K))$	55 2.16	95 1.26	125 1.1	105 1.23	100 1.22	65 1.84	110 1.23	100 1.22	85 3.5
	Medium Pressure Gas (20 bar)	$U (W/m^2 K)$ $C (£/(W/K))$	95 1.5	300 0.86	350 1	430 0.76	375 0.8	120 1.16	530 0.6	390 0.8	240 0.94
	High Pressure Gas (150 bar)	$U (W/m^2 K)$ $C (£/(W/K))$	120 2.05	350 1.1	400 1.1	500 1	400 1.1	150 1.3	600 1	420 1.1	350 1.2
	Treated Cooling Water	$U (W/m^2 K)$ $C (£/(W/K))$	105 1.4	484 0.75	500 1	940 0.5	715 0.72	145 1	1610 0.4	765 0.72	345 0.9
	Low Viscosity Hydrocarbon Liquid	$U (W/m^2 K)$ $C (£/(W/K))$	100 1.45	375 0.8	425 1.05	600 0.8	500 0.9	130 1.1	820 0.5	525 0.9	290 1.2
	High Viscosity Hydrocarbon Liquid	$U (W/m^2 K)$ $C (£/(W/K))$	70 2.4	140 1.66	175 1.2	160 0.95	155 1	85 2.7	175 0.82	155 1.6	125 1.95
	Boiling Water	$U (W/m^2 K)$ $C (£/(W/K))$	105 1.4	470 0.9	550 1	875 0.5	670 0.6	140 1.1	1435 0.4	725 0.73	340 0.9
	Boiling Organic Liquid	$U (W/m^2 K)$ $C (£/(W/K))$	100 1.45	375 0.8	430 1.05	600 0.8	500 0.9	130 1.28	820 0.7	525 1	285 1.1

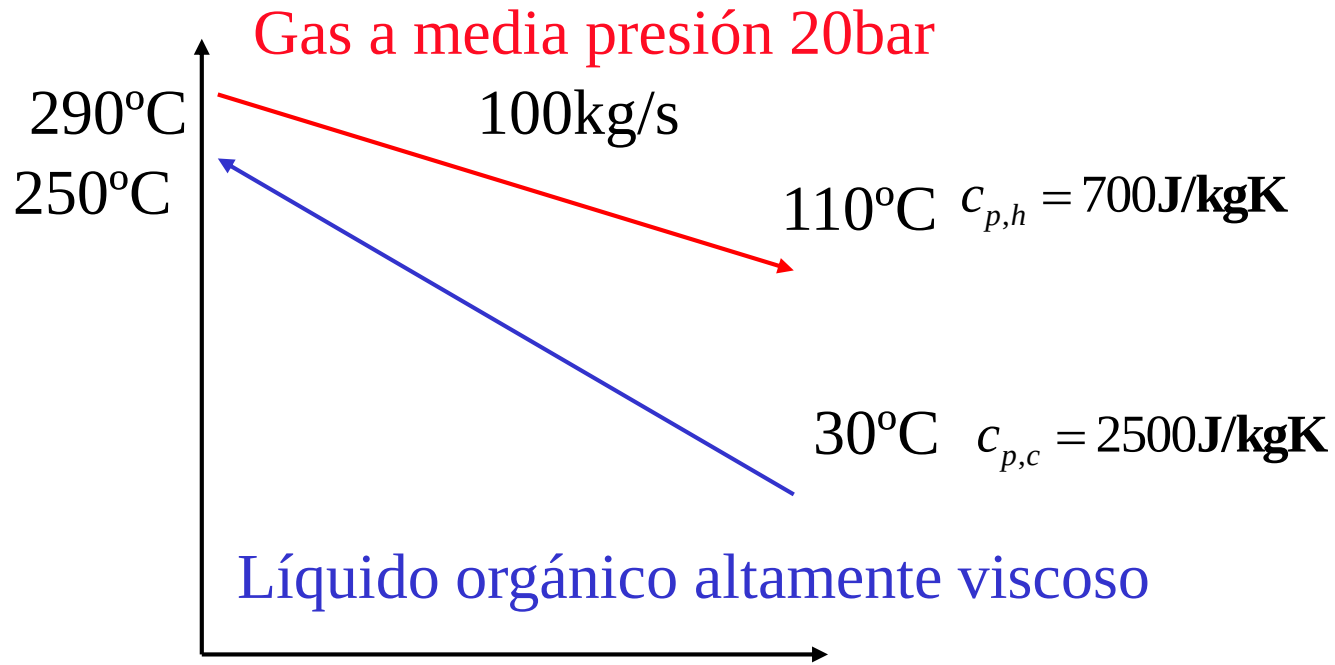
$$C_1 = 2.5; (Q/\Delta T_m)_1 = 1000$$

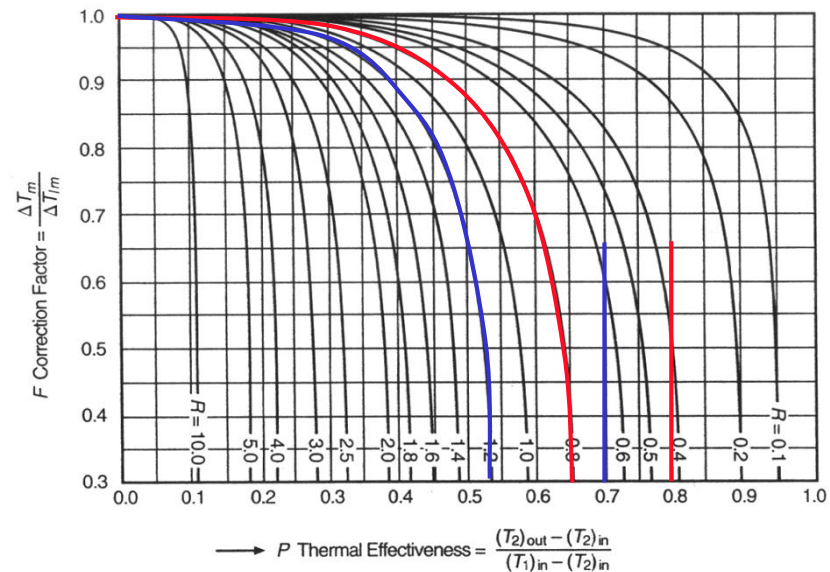
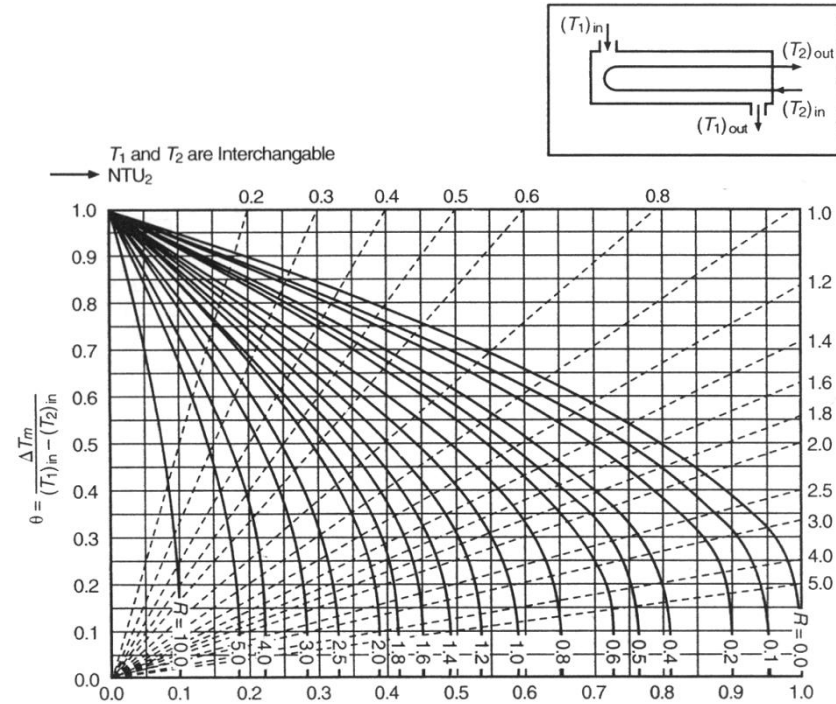
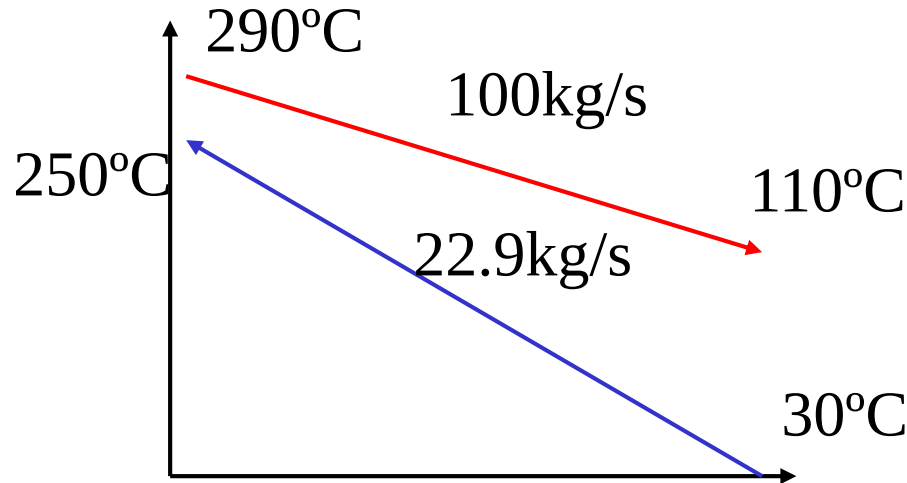
$$C_2 = 1; (Q/\Delta T_m)_2 = 5000$$

$$C = 1.84$$

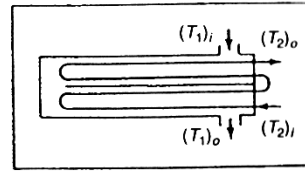
$$\text{Coste doble tubo} = 1.84 \cdot 1713 = 3152£$$

Determinar el número de carcasas necesarias y el coste del intercambiador para el intercambio propuesto en la figura. Teniendo en cuenta las modificaciones sobre el standard necesarias



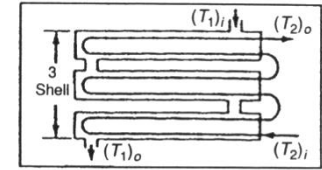
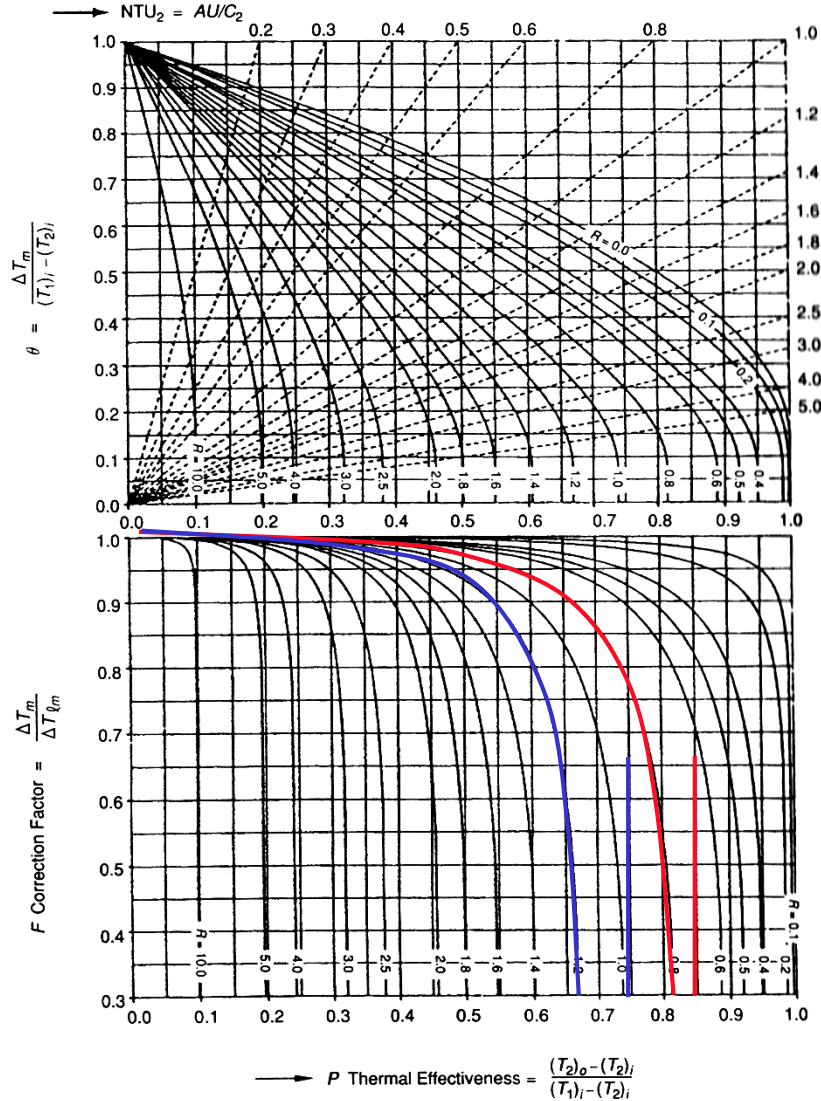


Coste de un tubo carcasa (Nº de carcasa)



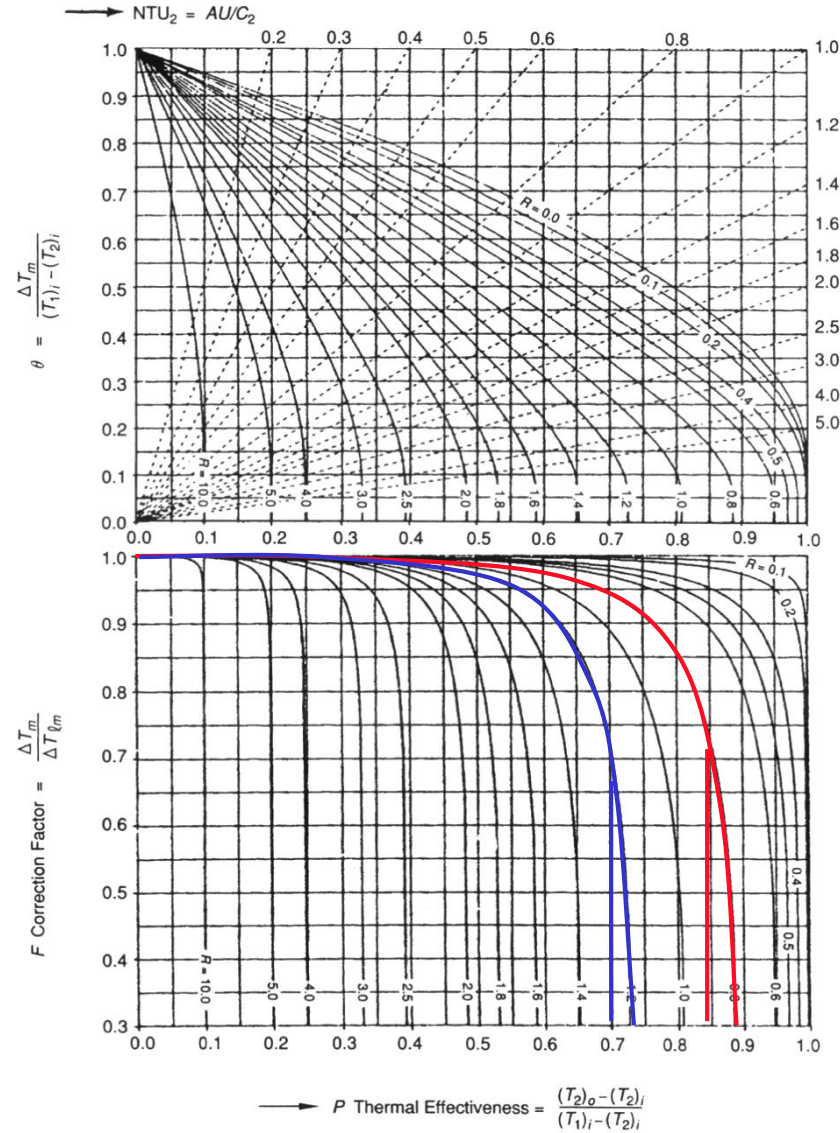
$$R = \frac{C_2}{C_1} = \frac{(T_{1i}) - (T_{1o})}{(T_{2o}) - (T_{2i})}$$

T_1 and T_2 are interchangeable



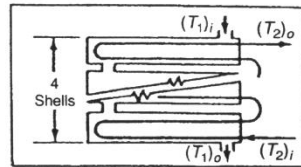
$$R = \frac{C_2}{C_1} = \frac{(T_{1i}) - (T_{1o})}{(T_{2o}) - (T_{2i})}$$

T_1 and T_2 are interchangeable



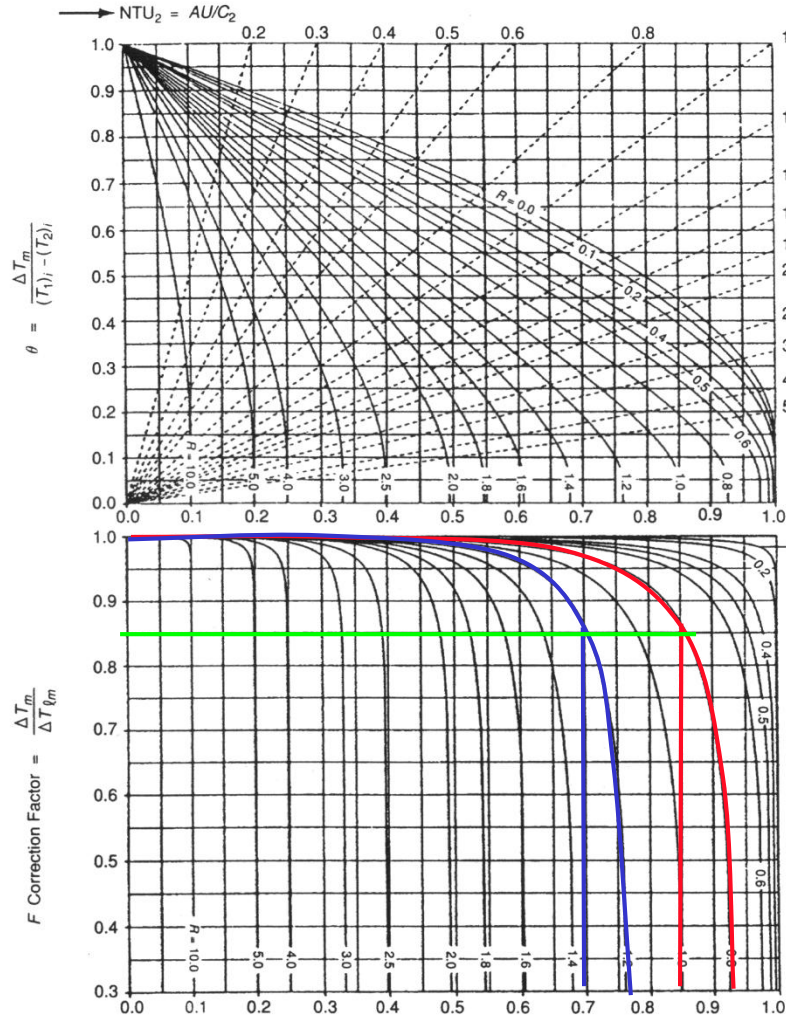
$$UA = 256862 \text{ W/m}^2\text{K}$$

$$UA/4 = 64215 \text{ W/m}^2\text{K}$$



$$R = \frac{C_2}{C_1} = \frac{(T_1)_i - (T_1)_o}{(T_2)_o - (T_2)_i}$$

T_1 and T_2 are interchangeable



$$P \text{ Thermal Effectiveness} = \frac{(T_2)_o - (T_2)_i}{(T_1)_i - (T_2)_i}$$

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 1.11	93 0.76	120 1.06	102 0.73	99 0.74	63 0.99	107 0.71	100 0.73	86 0.80
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	93 0.76	300 0.37	350 0.62	500 0.28	375 0.33	120 0.63	530 0.28	388 0.32	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 1.06	350 0.62	400 0.94	600 0.40	450 0.53	200 0.73	600 0.40	400 0.62	300 0.62
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.71	484 0.29	600 0.40	938 0.23	714 0.25	142 0.56	1607 0.19	764 0.24	345 0.34
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.38	130 0.59	818 0.24	524 0.28	286 0.38
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	68 0.94	138 0.57	200 0.73	161 0.52	153 0.53	82 0.83	173 0.50	155 0.53	124 0.60
	Boiling Water	U (W/m ² K) C (€/(W/K))	105 0.71	467 0.29	550 0.49	875 0.23	677 0.25	140 0.56	1432 0.20	722 0.25	336 0.35
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.28	130 0.59	818 0.24	524 0.28	286 0.38
	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (€/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (€/(W/K))	93 0.58	300 0.23	350 0.35	429 0.18	375 0.20	120 0.47	530 0.16	388 0.19	240 0.27
100 000	High Pressure Gas (150 bar)	U (W/m ² K) C (€/(W/K))	120 0.93	350 0.35	400 0.58	600 0.24	450 0.28	200 0.64	600 0.24	400 0.32	300 0.39
	Treated Cooling Water	U (W/m ² K) C (€/(W/K))	105 0.52	484 0.17	600 0.24	938 0.116	714 0.134	142 0.41	1607 0.086	764 0.129	345 0.21
	Low Viscosity Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.55	375 0.20	450 0.28	600 0.145	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	High Viscosity Liquid	U (W/m ² K) C (€/(W/K))	68 0.77	138 0.42	200 0.64	161 0.37	153 0.38	82 0.65	173 0.35	155 0.38	124 0.45
	Boiling Water	U (W/m ² K) C (€/(W/K))	105 0.52	467 0.168	550 0.26	875 0.121	677 0.137	140 0.41	1432 0.091	722 0.133	336 0.22
	Boiling Organic Liquid	U (W/m ² K) C (€/(W/K))	99 0.55	375 0.20	450 0.28	600 0.146	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24

$$C_1 = 0.57; (Q/\Delta T_m)_1 = 30000$$

$$C_2 = 0.42; (Q/\Delta T_m)_2 = 100000$$

$$A = 464 \text{ m}^2$$

$$C = 0.47 \frac{\text{£}}{\left(\frac{\text{W}}{\text{K}}\right)}$$

$$U = 138 \frac{\text{W}}{\text{m}^2\text{K}}$$

$$\text{Coste} = 0.47 \cdot 64000 \cdot 4 = 120730 \text{ £}$$